

Technology Stack

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Category	Technology / Tool
Programming Language	Python
Data Handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-Learn (Logistic Regression, Random Forest, Decision Tree), XGBoost
Model Persistence	Pickle / Joblib
Web Framework	Flask
Frontend	HTML, CSS, Jinja2 Templates
Cloud Deployment	IBM Watson Machine Learning
IDE	Jupyter Notebook, VS Code
Version Control	Git, GitHub

Skills Required

XgboostMachine Learning AlgorithmsArtificial IntelligenceDecision Tree LearningNumPy (Python Package)

Python (Programming Language)Scikit-Learn (Python Package)Matplotlib (Python Package)Flask (Python Framework)

Architecture Layer Mapping

Layer	Technology
Presentation Layer	Flask templates (HTML/CSS) rendering the input form and prediction result
Application / Logic Layer	Flask app routes handling requests, calling preprocessing and model inference functions
Model Layer	Trained classification model (best of Logistic Regression, Random Forest, XGBoost, Decision Tree)
Data Layer	Historical applicant dataset used for training; incoming applicant data for inference
Deployment Layer	IBM Watson Machine Learning for cloud hosting and scalable real-time predictions